

M/B — KIGALI, RWANDA

Mucyo Bertrand.

- WEB & MOBILE ENGINEER
- BACKEND ENGINEER
- APPLIED AI ENGINEER

PROFILE

Engineer and product lead who turns messy, real-world problems into software people actually use. I build **web and mobile apps**, **production backends** and **applied AI**. I lead product for four AI products at **Entelon** and build community health and livelihood tools at **IWP Africa**, all while studying at **Rwanda Coding Academy**, where I was admitted through a competitive national selection.

HIGHLIGHTS

RCA **Admitted to Rwanda Coding Academy (2024)**: Africa's first coding-focused high school, through a merit-based national selection of top STEM students in Mathematics and Physics, with around 120 places a year.

04 **AI products led** at Entelon across education, travel, field research and call centres.

07 **Products shipped or in progress** across Entelon and IWP Africa in AI, health, education and agriculture.

25+ **Public repositories** spanning web, mobile, backend, IoT and ML.

EXPERIENCE

Product Lead @ Entelon

PRESENT

AI company, Kigali · AI products and custom AI systems that save businesses and institutions time and cost

- Lead product for Entelon's AI portfolio: **Klok AI**, **HeyRwanda**, **Visko** and **Phibram**, from discovery and roadmap to launch.
- **Klok AI (education):** reads students' handwritten exam answers and suggests marks step by step, with the teacher reviewing and owning every grade.
- **Visko (field research):** teams keep their existing questionnaires and speak the answers, so data moves to analysis with no manual entry.
- **Phibram (call centres):** listens to every call and turns hotline conversations into a continuous view of service quality and customer experience.
- **HeyRwanda (travel):** a companion for choosing where to go, finding your way and paying along the trip.
- Work across Entelon's delivery cycle (workflow analysis, integration, ongoing maintenance) on solutions such as workflow automation, voice agents, custom chatbots and AI data analysis.

Developer & Researcher @ IWP Africa

PRESENT

Insight Wellbeing Partners · community-first, evidence-led health, wellbeing and livelihood solutions

- **Mamora (health):** voice-based platform where households report symptoms as they appear, so community health workers are informed immediately and outbreaks are detected early.
- **ECD Intelligence (early childhood):** plain-language daily guidance for ECD centre managers on each child, adapted weekly from simple health and development measurements.
- **Trace (agriculture):** record book for farmers and marketplace for buyers; every product carries its full farm history.
- Pair engineering with field research and community data systems, following IWP's **Listen, Act, Measure, Influence** approach so evidence feeds back into programme design.

CONTACT

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GITHUB
github.com/mucyobertrand

BASED IN
Kigali, Rwanda

EDUCATION

Rwanda Coding Academy

2024 — 2027 · NYABIHU, WESTERN PROVINCE

Admitted on merit through a rigorous national selection: exams and assessments of problem-solving and coding aptitude.

Three-year, fully practical programme covering **software engineering**, **embedded systems**, **cybersecurity** and **artificial intelligence**.

CORE SKILLS

Web & Mobile Engineering

Next.js and React web apps; cross-platform mobile with React Native and Expo; responsive, animated UI.

Backend Engineering

NestJS and Node.js REST APIs on PostgreSQL with Prisma: OTP auth, payments, mobile-money and SMS integrations, Swagger docs.

Applied AI Engineering

LLM and voice agents, handwriting and document understanding, retrieval over private data, evaluation.

Product & Research

Discovery, roadmaps and launches; field research, community data and evidence-led iteration.

HOW I WORK

01 **Listen** to the people closest to the problem.

02 **Build** the smallest useful thing, fast.

03 **Measure** usage and outcomes.

04 **Improve** from evidence, not assumptions.

EXPERIENCE (CONTINUED)

Independent Engineer @ Open source

2023 — PRESENT

25+ public repositories · github.com/mucyobertrand

- Design and ship full products end to end: data model, API, web or mobile client, and deployment.
- Build NestJS backends documented with **Swagger/OpenAPI**, with OTP authentication and PostgreSQL via Prisma.
- Prototype IoT and embedded builds with **ESP8266**, **RFID** and **MQTT**, connected to real-time web dashboards.

SELECTED PROJECTS

Velo

NESTJS · PRISMA

- Personal-finance app that reads MTN & Airtel mobile-money SMS **on-device**, keeping raw messages private.
- Built the API: phone-number **OTP login** (handles 07..., +250..., 00250... formats), bearer-token auth, premium unlock codes and signed reports.
- Parsing rules served from the API and seeded per network; Swagger docs and admin-key routes.

github.com/mucyobertrand/velo-app-backend

Dent-Wise

NEXT.JS 15 · POSTGRES

- Dental practice platform: patients book appointments, manage records and message dentists.
- Next.js 15 App Router, TypeScript, Tailwind CSS 4, Radix + shadcn/ui, React Query.
- Clerk authentication; PostgreSQL with Prisma ORM; doctor management for clinics.

github.com/mucyobertrand/dent-wise

Career Space

REACT NATIVE · TS

- Expo (React Native) career app with its own TypeScript backend service.
- Built end to end, from data model and API to mobile UI.

github.com/mucyobertrand/career-space

PRODUCT PORTFOLIO

- **Entelon:** Klok AI (education) · HeyRwanda (travel & payments) · Visko (voice field research) · Phibram (call-centre analytics).
- **IWP Africa:** Mamora (community health reporting) · ECD Intelligence (early childhood) · Trace (traceable produce marketplace).

TECHNICAL TOOLKIT

LANGUAGES

TYPESCRIPT · JAVASCRIPT · PYTHON · SQL · HTML/CSS

WEB & MOBILE

REACT · NEXT.JS · REACT NATIVE · EXPO · TAILWIND · SHADCN/UI · REACT QUERY · MOTION

BACKEND & DATA

NODE.JS · NESTJS · REST · POSTGRESQL · PRISMA · SWAGGER · OTP AUTH · CLERK

APPLIED AI

LLMS · VOICE AI · HANDWRITING / OCR · RAG · ML IN PYTHON

IOT & EMBEDDED

ESP8266 · RFID · MQTT

TOOLS

GIT · GITHUB · VERCEL

DOMAINS

AI · HEALTH · EDUCATION · AGRICULTURE · FINTECH · TRAVEL

> “Build technology the people closest to the problem actually want to use, and prove it works.